

Automation Developing Kit (AD Kit)



Distributed/Integrated Modules with Powerful Graphical Object Oriented (GOO) SCADA solution

Provided by

HALTK

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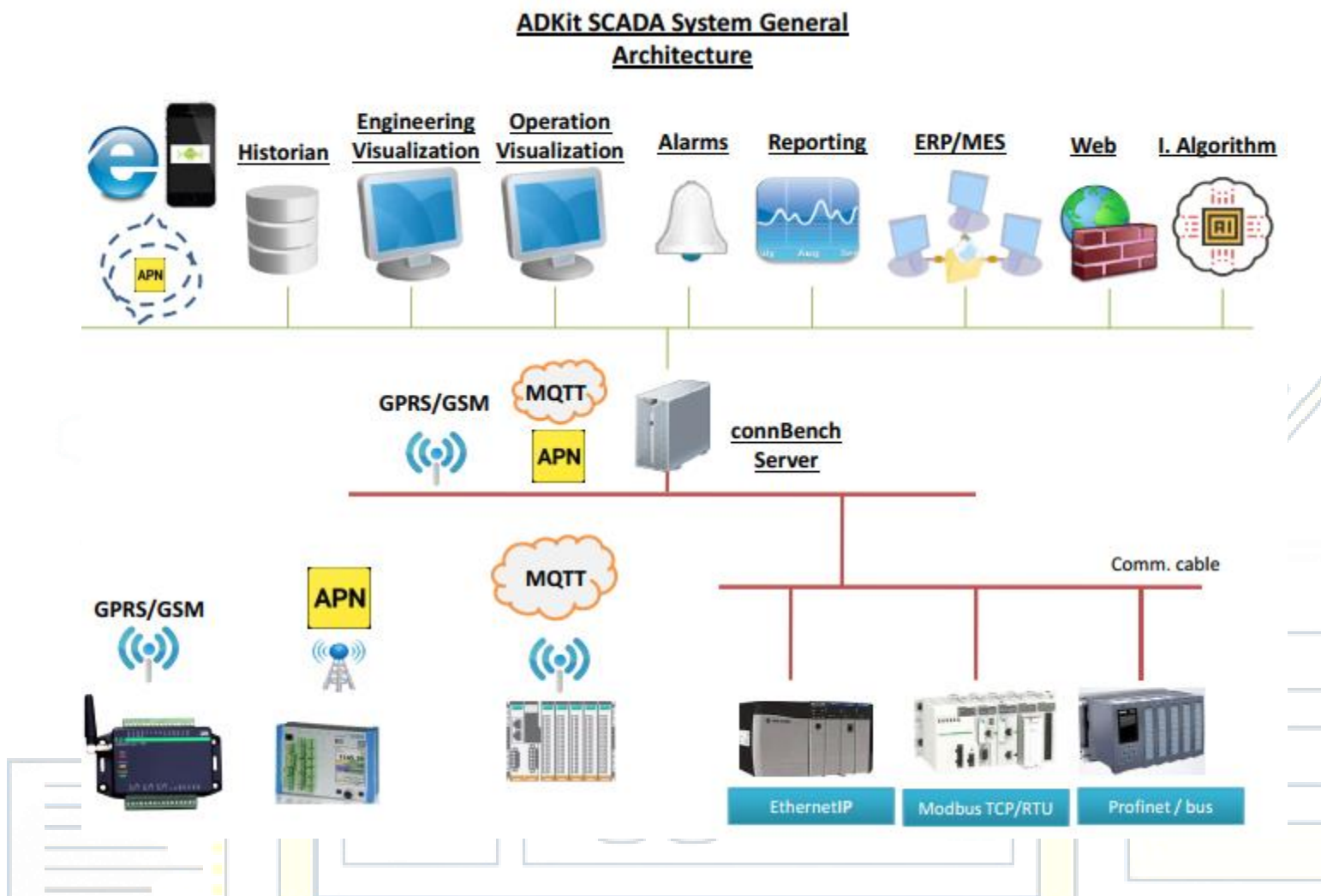
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Automation Developing Kit (AD Kit)

Distributed/Integrated Modules with Powerful Graphical Object Oriented (GOO) SCADA solution



AD Kit represents family of the Modern SCADA Systems which supports the technology of Fully Integrated Standalone Modules, All modules are hot-pluggable where any module can be installed, updated, removed separately without affecting the operation of the system operation, and all modules work together seamlessly and can be used to create any kind of industrial automation solution, including SCADA visualization, historian and remote mobile monitoring and beyond, To enable users adding features such as trends, tables, mobile access, SMS notifications for alarms, transaction management, sophisticated logic systems, enterprise administration, and much more.

AD Kit is robust and reliable solution in a multi modular software packages that scales from a stand-alone HMI system to a distributed solution.

AD Kit suitable for plant process, batch and discrete applications with the realtime data that provides a complete industrial automation solution.

AD Kit provides easy and user visual ability of objects animation and configuration without writing any code based on the technology of Graphical Object Orientation (GOO) which will give designer about 60% reduction in engineering effort and time to create applications due to ease of use.

AD Kit Interfaces with many field devices and manufacturing plant information system without need for any 3rd part OPC / IO driver.

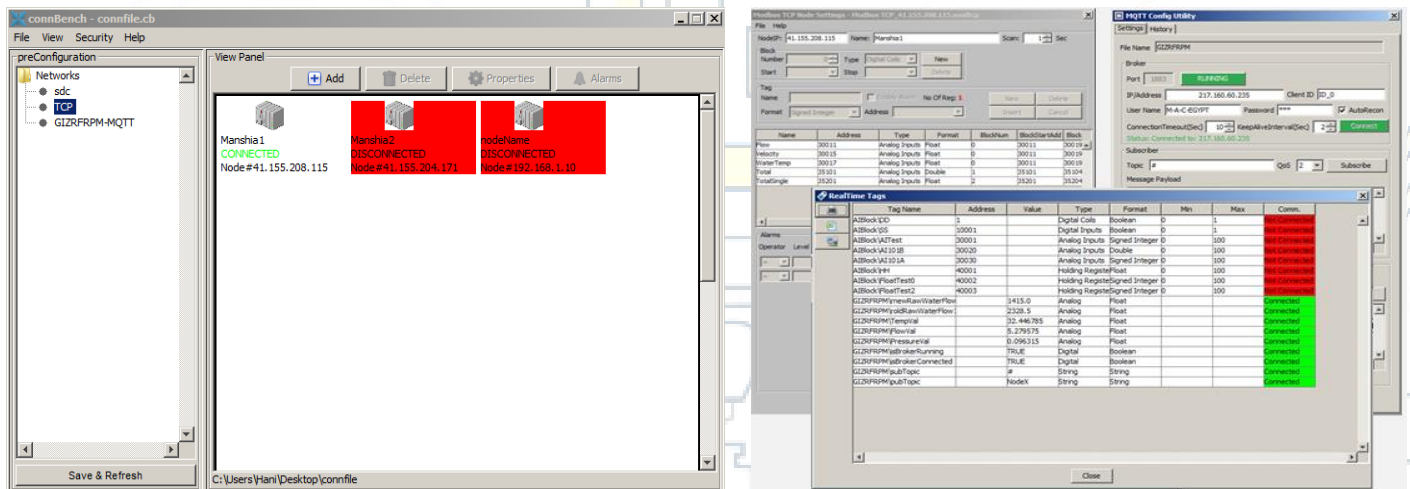
AD Kit supply operators with responsive visualization, alarming, flexible analysis and powerful reporting tools.

AD Kit SCADA System contains at least the below modules

- **Communication Module (connBench Communication Server)**

Connect to most major programmable logic controllers (PLCs). This module serves as a totally cross-platform Server & Clients

Communication Module provides both server and client functionality. As a server, the module integrates with popular drivers for Allen-Bradley EthernetIP, Siemens Proginet, Modbus TCP/RTU, GPRS, SMS, MQTT devices. As a client, the module has the ability to make outgoing connections to other SCADA system modules.



connBench Data Security

All module integration based on multi-threaded TCP protocol

AD Kit SCADA System has the ability to add a Secure Client Authentication, permissions model enables role-based security for complete system protection.

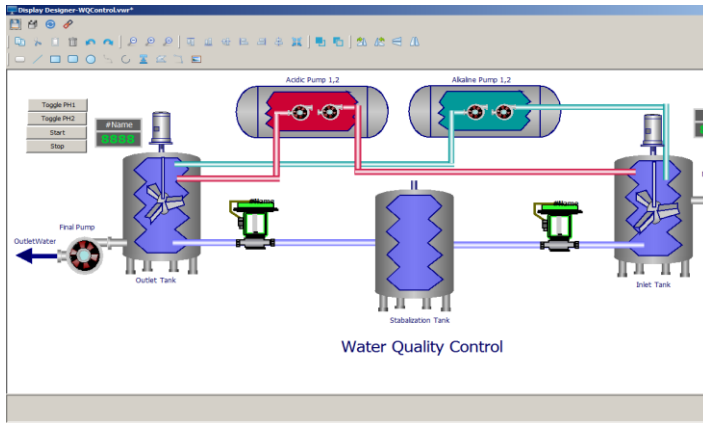
- **Visualization Module**

The visualization offers complete supervisory control, process monitoring and operator interface software for the Windows and Windows Server with small footprint for operating systems.

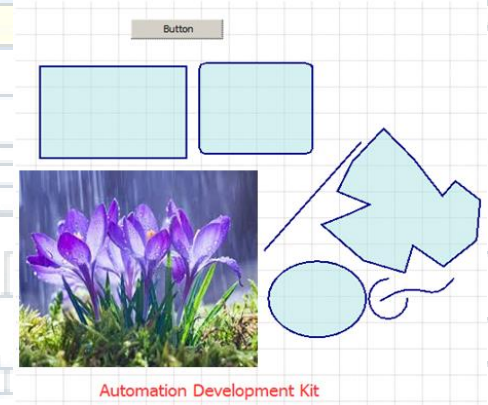
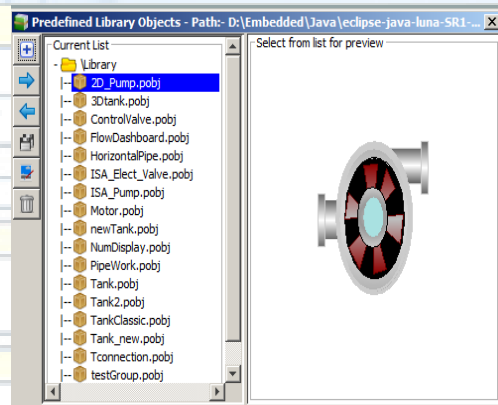
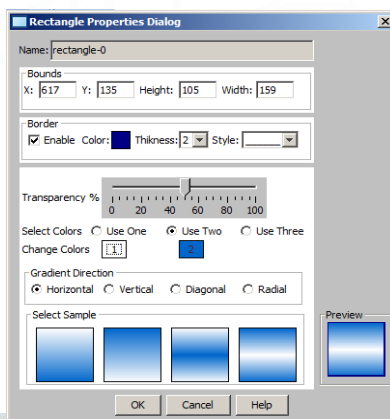
The visualization module for plant-floor and desktop screens Display charts, graphs, trends, and more.

The visualization module for mobile-responsive industrial applications easily builds visually application for monitoring and control using HTML for any size of screen.

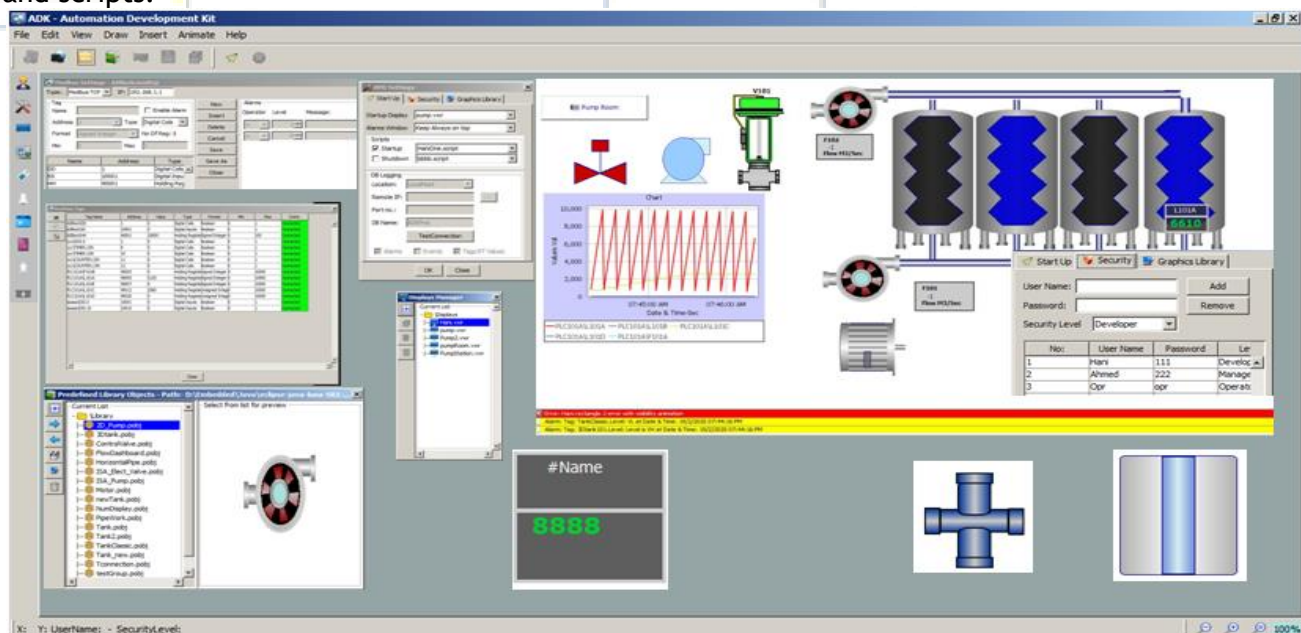
The visualization module uses the click and builds graphical object-based programming approach to completely eliminate programming and scripting.



Symbols/Library: Included library features push buttons, pilot lights, tanks, sliders, meters, motors, pipes, valves and other common objects. User can modify existing symbols to suit his needs, or create his own from scratch, Change in library object (parent object) will take effect to all child objects in the project based on Goo technology.



Graphics and Design Tools: Create screens to meet any application requirement using the tools in our graphic interface to create any functionality required. Store graphics in the library for future use, and easily make projects across a product line share. Graphics library can be imported / exported to be used with future projects. Graphics library exported with all the parent graphics with tags, alarms, animations and scripts.



Animation: Take command over graphics in a user friendly and intuitive interface. Import images, and even rotate dynamically using custom rotation points. Fill bar graphs with color, or adjust the scale of objects with easy-to-use configuration. Other animations include 'command' (for touch, keyboard and mouse interaction), text data, color, resize, transparency, and position.

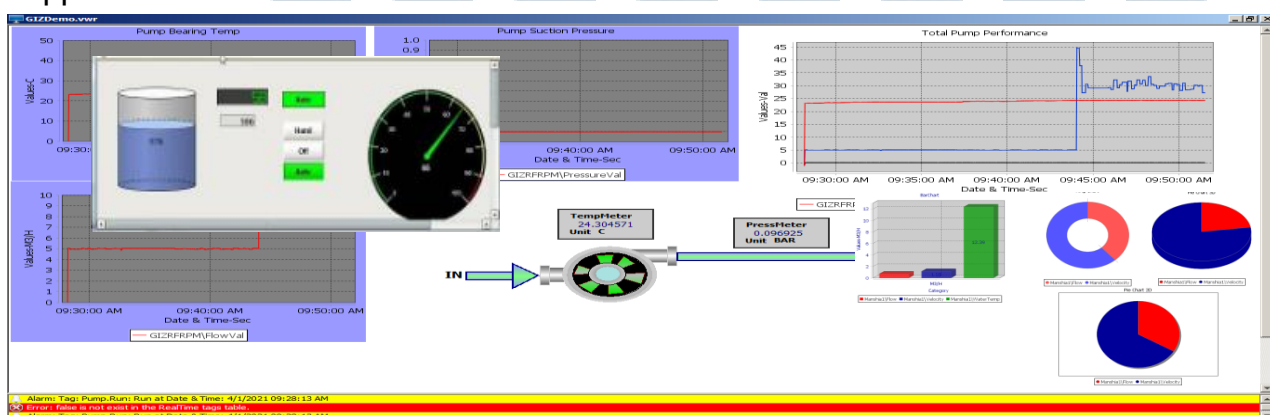
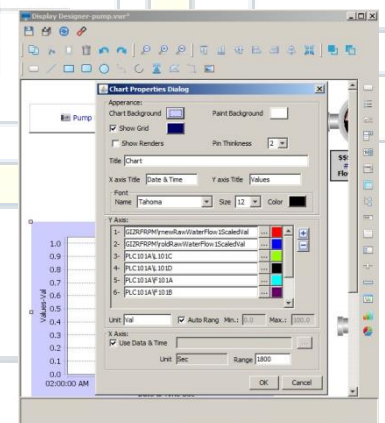
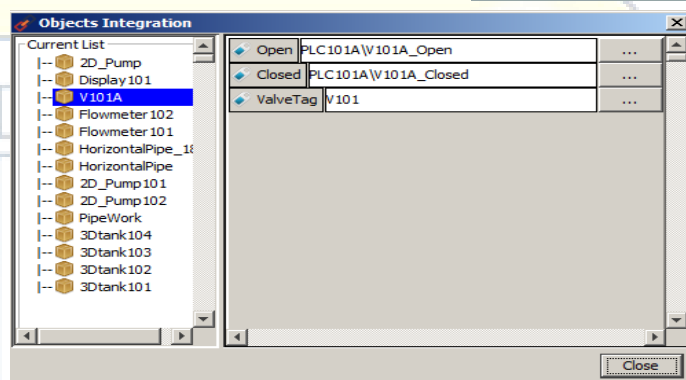
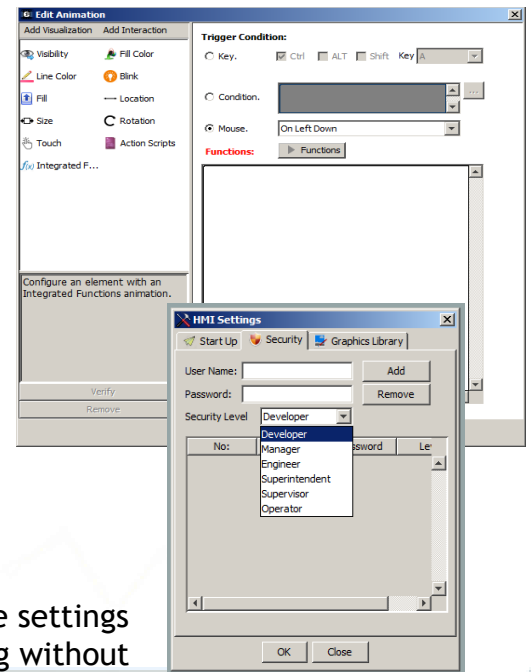
Security: Visualization Module includes access support for group and user accounts by setting user name and password with authorization level (Manager, Developer, Engineer, Supervisor & Operator).

Tag Database: create, delete, or modify the tags database settings during online/offline with Alarm configuration for each tag without need to interrupt the process operation at connBecn server while visualization module will update its tags database automatically.

Tag database will be integrated with each child object local tags during offline for animation & interactions.

Trends: Visualization Module Provides support for various Pie Charts and XY Charts. ... The trends that are represented on an XY-plane include Area Chart, Bar Chart, Bubble Chart, Line Chart, Scatter Chart, Stacked Area Chart, Stacked Bar Chart, etc. Each chart is represented by a class and all these charts belong to the package AD Kit.

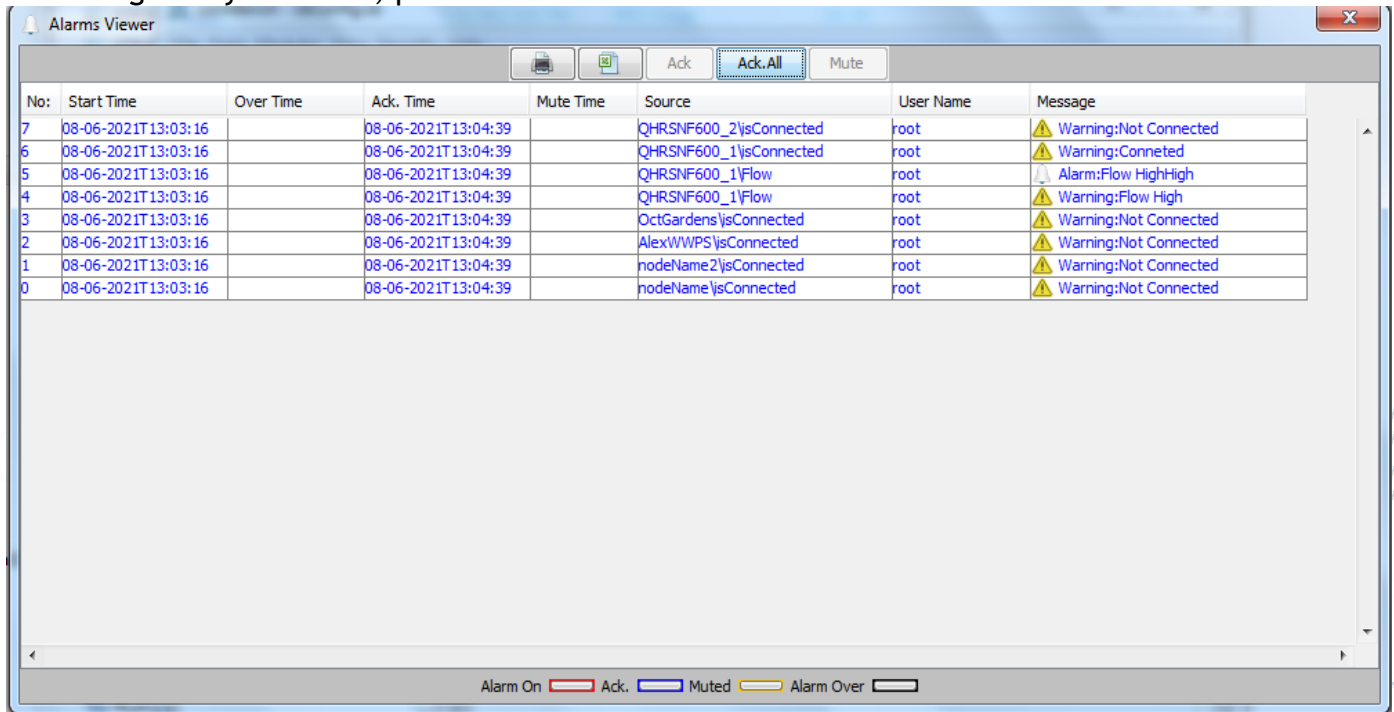
Real-time and Historical trends are supported for all kinds of charts. Color or fill trends with graphic elements to enhance clarity of data. Date/Time based or numeric (X/Y plot) trends give operator the flexibility to display information that best suits the application.



- Alarm Notification Module

Configure the logic for how, why, and when alarm notifications are delivered, manage alarm notification for groups of users, and send notifications via email.

Alarms: Can send online alarms or reports using standard formats like PDF/Excel. Alarms are real-time and historical; log data to a database. Use remote notification to send alarms right to your inbox, printer.



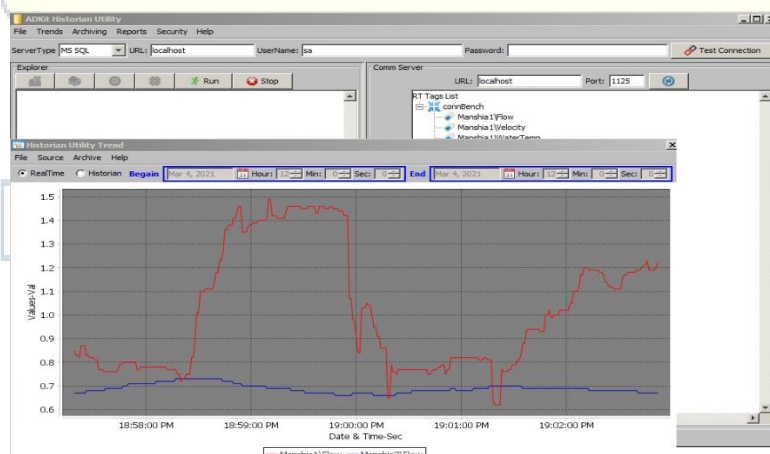
The screenshot shows the 'Alarms Viewer' window with a table of alarm events. The table has columns: No., Start Time, Over Time, Ack. Time, Mute Time, Source, User Name, and Message. The messages include warnings like 'Warning:Not Connected', 'Warning:Conncted', 'Alarm:Flow HighHigh', and 'Warning:Flow High'. Below the table, there is a legend for alarm states: Alarm On (red bar), Ack. (blue bar), Muted (yellow bar), and Alarm Over (grey bar).

No.	Start Time	Over Time	Ack. Time	Mute Time	Source	User Name	Message
7	08-06-2021T13:03:16		08-06-2021T13:04:39		QHRSNF600_2\jsConnected	root	Warning:Not Connected
6	08-06-2021T13:03:16		08-06-2021T13:04:39		QHRSNF600_1\jsConnected	root	Warning:Conncted
5	08-06-2021T13:03:16		08-06-2021T13:04:39		QHRSNF600_1\Flow	root	Alarm:Flow HighHigh
4	08-06-2021T13:03:16		08-06-2021T13:04:39		QHRSNF600_1\Flow	root	Warning:Flow High
3	08-06-2021T13:03:16		08-06-2021T13:04:39		OctGardens\jsConnected	root	Warning:Not Connected
2	08-06-2021T13:03:16		08-06-2021T13:04:39		AlexWWPS\jsConnected	root	Warning:Not Connected
1	08-06-2021T13:03:16		08-06-2021T13:04:39		nodeName2\jsConnected	root	Warning:Not Connected
0	08-06-2021T13:03:16		08-06-2021T13:04:39		nodeName\jsConnected	root	Warning:Not Connected

- Historian Database Module

Database module will be based on using unlimited size database engine and user-friendly historian configuration interface for easily log data, call stored procedures.

Database: Can connect to SQL database (Microsoft SQL, MySQL), or Microsoft Access or Excel or PostgreSQL, without previous require knowledge of SQL language.



- Reporting Module:

The reporting shall be based on business intelligent reporting & charting technology with ability to end user to design / modify reports without need to call our service, where end user can create dynamic, database-driven PDF, Excel, Word reports with pixel-perfect layout with ease where the **Historian** module can select Data from Many

localhost | ReportServer/Pages/ReportViewer.aspx?%2fG2Report+Project%2fGeneralReports+Command-Render

Change Credentials

Project Report accessed from the web browser and it can be uploaded to web host for online anywhere access

View Report

AdKIR HMI Solutions

General Report

6th October Booster PS Flow

Pulse Title

Axis Title

New Raw Water Flow/Scanned Val
Old Raw Water Flow/Scanned Val

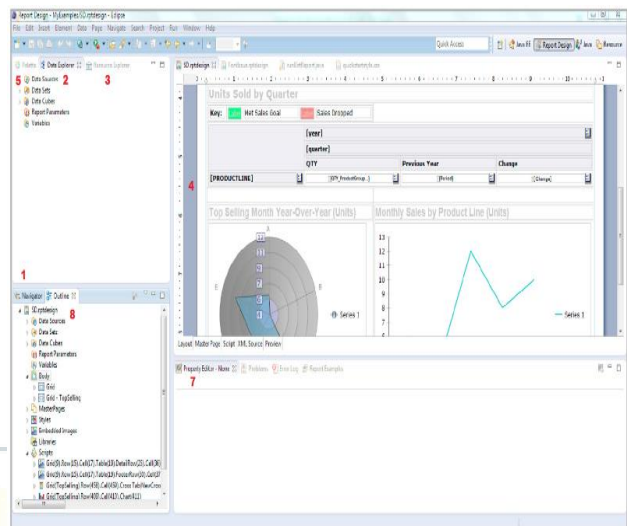
TestBig Main Pump Performance Chart

Pulse Title

Axis Title

Temp Wk
Flow Wk
Pressure Wk

DATE\END TIME	Raw Water Flow1	Raw Water Flow2	DATE\END TIME	Temperature (C)	Water Flow (M3/Hr)	Pressure (BAR)
12/8/2020 2:13:13 PM	1117	2207.5	12/8/2020 2:13:13 PM	35.13848	5.26302	0.096071
12/8/2020 2:13:23 PM	1225	2247.5	12/8/2020 2:13:23 PM	35.15941	5.26857	0.095304
12/8/2020 2:13:33 PM	1223	2218.5	12/8/2020 2:13:33 PM	35.15941	5.25744	0.095949
12/8/2020 2:13:43 PM	1221	2183.5	12/8/2020 2:13:43 PM	35.15679	5.25744	0.095949

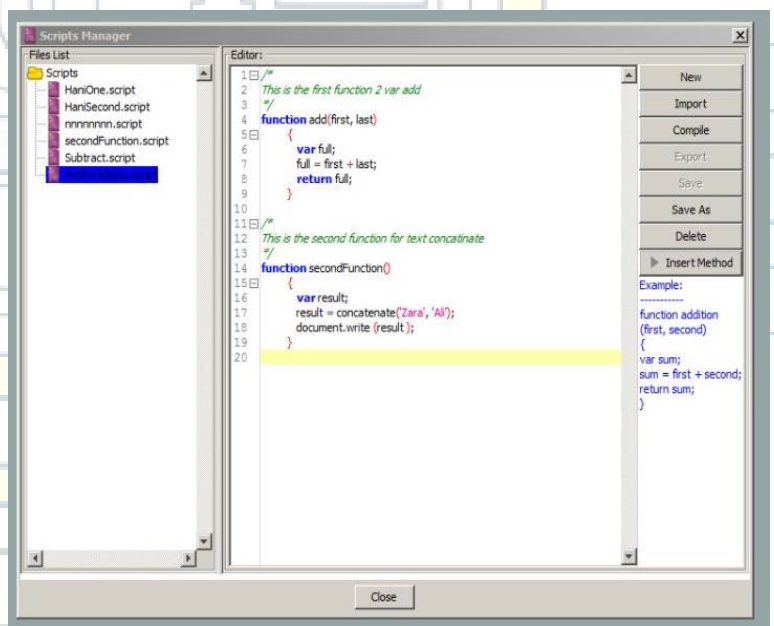


PDF Export: Can Generate Alarms, Reports, or any Graphics screen to be shared with the production, managers, or maintenance staff using the PDF format.

AD Kit proud to be one of the first SCADA systems that apply the AI scripting modules, where in machine learning, an algorithm is a set of rules given to an AI program to help it learn on its own. Traditional algorithms range from simple business rules to highly complex decision engines that require greater involvement of data scientists in tuning, maintenance and re-calibration. As a result, they afford greater transparency and control than AI that runs in auto-pilot mode.”

However, with more control comes a higher degree of responsibility, “The programmer must include all the rules and regulations for the algorithm to work properly because it has no common sense and no idea of things that are obviously wrong to us because the program does not understand it. the algorithm needs to be perfect with very specific and clear action plans to properly work.”

Standard Java Script is a powerful scripting language that **AD Kit** support. Take advantage of widely available resources for Java Script that can be used to give the user the functionality he need.

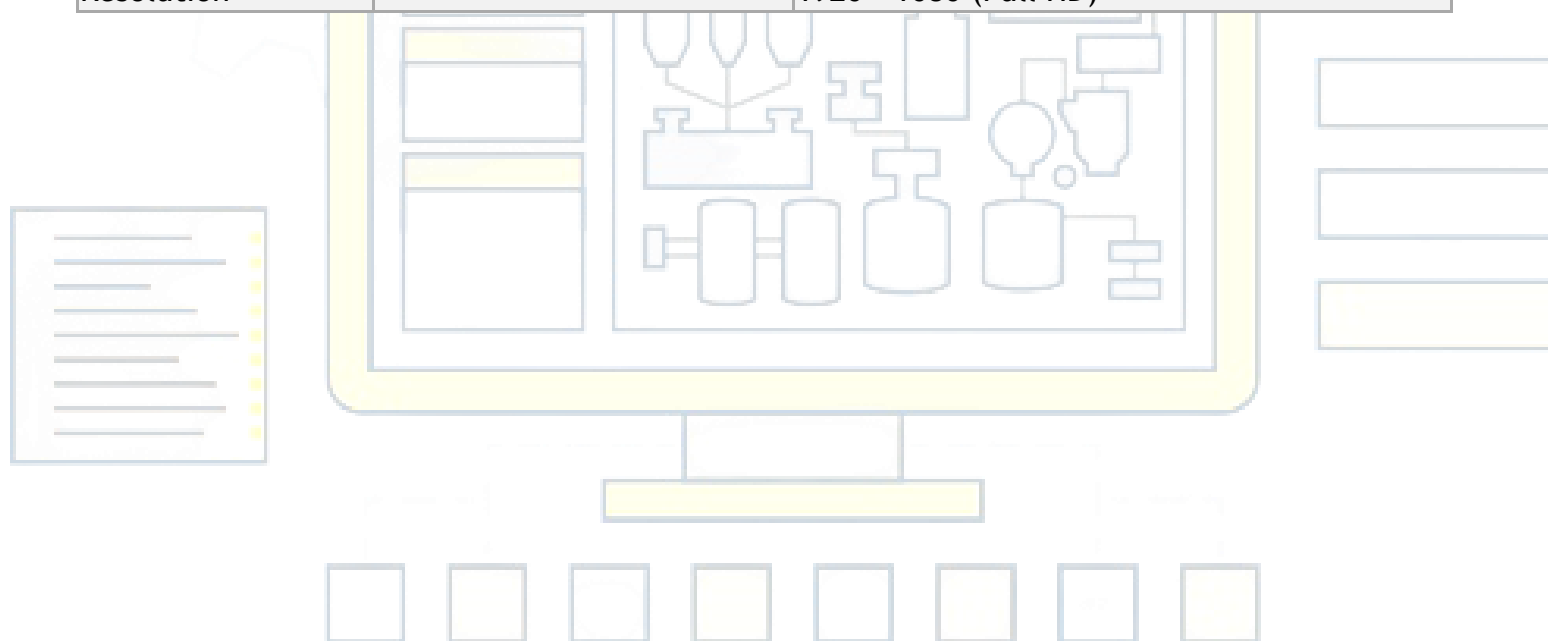


- Technical Specification

Certain hardware configuration conditions must be fulfilled for installation.

To efficiently work with AD Kit, select a system with the recommended specifications.

		MIN Recommended
CPU	Windows 10/7(64-32bit)	Multi core CPU Client: 3 GHz
	Windows Server 2012 R2 / Windows Server 2016	Multi core CPU Single-user system / server: 3.5 GHz
Work memory	Windows 10/7(64-32bit)	8 GB
	Windows Server 2012 R2 / Windows Server 2016	16 GB
Free storage space on the hard disk		Installation: Client: > 1.5 GB Server: 2 GB Working with: Client: > 1.5 GB Server: 10 GB
Color depth		Highest (32 Bit)
Resolution		1920 * 1080 (Full HD)



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